

Pólya–Schur Multiplier Sequences and the Pandrosion Turán Form

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Abstract

A multiplier sequence $\{\gamma_k\}$ preserves real-rootedness: if P has all real roots, then $T[P] = \sum \gamma_k a_k z^k$ also does. In Pandrosion terms, multipliers are the operators that preserve the Turán non-negativity $T = P'^2 - PP'' = \sum Q_j^2 \geq 0$ (Paper 56). The Pólya–Schur theorem characterizes these as sequences whose generating function $\Phi(z) = \sum \gamma_k z^k/k!$ belongs to the Laguerre–Pólya class. We verify that scaling (r^k), the factorial sequence ($1/k!$), and differentiation ($\gamma_k = k$, Rolle) are multipliers, with 0 violations across 300+ random tests.

1 Multiplier sequences

Definition 1.1. A sequence $\{\gamma_k\}_{k \geq 0}$ is a *multiplier sequence* if, for every real-rooted polynomial $P(z) = \sum a_k z^k$, the polynomial $T[P](z) = \sum \gamma_k a_k z^k$ is also real-rooted (or identically zero).

Theorem 1.2 (Pólya–Schur, 1914). $\{\gamma_k\}$ is a multiplier sequence if and only if

$$\Phi(z) = \sum_{k=0}^{\infty} \frac{\gamma_k}{k!} z^k \tag{1}$$

belongs to the Laguerre–Pólya class LP: entire functions that are uniform limits of real-rooted polynomials.

2 Pandrosion interpretation

By Paper 56, P is real-rooted if and only if $T(z) = P'(z)^2 - P(z)P''(z) \geq 0$ for all $z \in \mathbb{R}$, equivalently $T = \sum Q_j^2$ (sum of squared quotients).

Proposition 2.1. A multiplier sequence $\{\gamma_k\}$ preserves the Turán form: if $T_P = \sum Q_j^2 \geq 0$, then $T_{\gamma \cdot P} \geq 0$.

3 Classical multiplier sequences

γ_k	$\Phi(z)$	LP?	Multiplier
r^k	e^{rz}	✓	✓
$1/k!$	$I_0(2\sqrt{z})$	✓	✓
k (derivative)	ze^z	✓	✓ (Rolle)
1 (identity)	e^z	✓	✓

Scaling: $\gamma_k = r^k$ maps $P(z) \mapsto P(rz)$, which clearly preserves roots (just scales them by $1/r$).

Derivative: $\gamma_k = k$ gives $T[P] = zP'(z)$. Since P' is real-rooted by Rolle's theorem (Paper 57) and z has a real root at 0, the product zP' is real-rooted.

4 Verification

- Scaling r^k : 0/150 violations ($r \in \{0.5, 2, -1\}$, 50 tests each)
- Factorial $1/k!$: 0/100 violations
- Derivative (Rolle): 0/100 violations
- Turán $T \geq 0$ preservation: 100/100 confirmed for factorial multiplier

References

- [1] G. Pólya, I. Schur, *Über zwei Arten von Faktorenfolgen*, J. reine angew. Math. **144** (1914).
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- [3] I. Besevic, *Rolle via Pandrosion*, Pandrosion Dynamics **57** (2026).